

Functional and differential equations

— 3

Qualitative theory: answers without solving the equation

Homework

13 сентября 2026 года

Theoretical minimum. Existence is Peano, uniqueness is Picard, and a solution lives until it leaves every compact set — so to prove global existence one needs an a priori bound, usually from Gronwall (Differentiable functions – 6). For an autonomous equation on the line the zeros of the right-hand side organise everything: motion is monotone between them, every bounded solution tends to an equilibrium, and there are no periodic solutions. In the plane trajectories cannot cross, a first integral confines them to level sets, and a Lyapunov function decides stability without any formula. The five questions worth asking about a differential equation are: does the solution exist globally, is it bounded, does it have a limit, does it oscillate, and does it return.

1. Prove that every solution of $y' = -y + g(x)$ with bounded continuous g is bounded on $[0, \infty)$, and that if $g(x) \rightarrow L$ then $y(x) \rightarrow L$; deduce the same for $y' = -y^3 + g$.
2. Prove that every solution of $y'' + q(x)y = 0$ with $q(x) \geq c > 0$ has infinitely many zeros, and that with $q(x) \leq 0$ it has at most one; compare with Sturm's theorem in Differentiable functions – 6.
3. Let f be continuous with $xf(x) > 0$ for $x \neq 0$. Prove that the equilibrium $x = 0$ of $\ddot{x} + f(x) = 0$ is stable, by constructing the energy as a Lyapunov function, and decide when it is asymptotically stable.
4. Prove that a bounded solution of $\dot{x} = f(x)$ on $\mathbb{R}$ with continuous f tends to a zero of f as $t \rightarrow +\infty$, and give an example in $\mathbb{R}^2$ where a bounded trajectory has no limit.

5. Prove that the Lotka–Volterra system $\dot{x} = x(a - by)$, $\dot{y} = y(cx - d)$ with positive parameters has the first integral $cx - d \ln x + by - a \ln y$ and that all trajectories in the open quadrant are closed curves.
6. Prove that the solution of $y' = y^2 - x$ with $y(0) = 0$ exists for all $x > 0$ and find its asymptotics as $x \rightarrow +\infty$; compare with problem 9 in the homework of Differentiable functions – 6.
7. Let A be a real $n \times n$ matrix. Prove that all solutions of $\dot{x} = Ax$ tend to zero if and only if all eigenvalues of A have negative real part, and that they are all bounded if and only if the eigenvalues with zero real part are semisimple (Linear algebra – 3).
8. Prove Gronwall's inequality in the integral form and use it to show that the solutions of $\dot{x} = f(t, x)$ with $|f(t, x)| \leq a(t)|x| + b(t)$ and integrable a, b exist on the whole interval.
9. Prove that a periodic solution of a planar autonomous system encloses an equilibrium, and give an example showing that in $\mathbb{R}^3$ a periodic trajectory can enclose nothing.
10. Let u be a non-negative solution of $u' = u \ln \frac{1}{u}$ on $(0, \infty)$ with $u(0) = u_0 \in (0, 1)$. Find the explicit solution, determine whether it reaches zero in finite time, and explain the connection with uniqueness.